This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2025.3549850

# Predicting and Explaining Cognitive Load, Attention, and Working Memory in Virtual Multitasking

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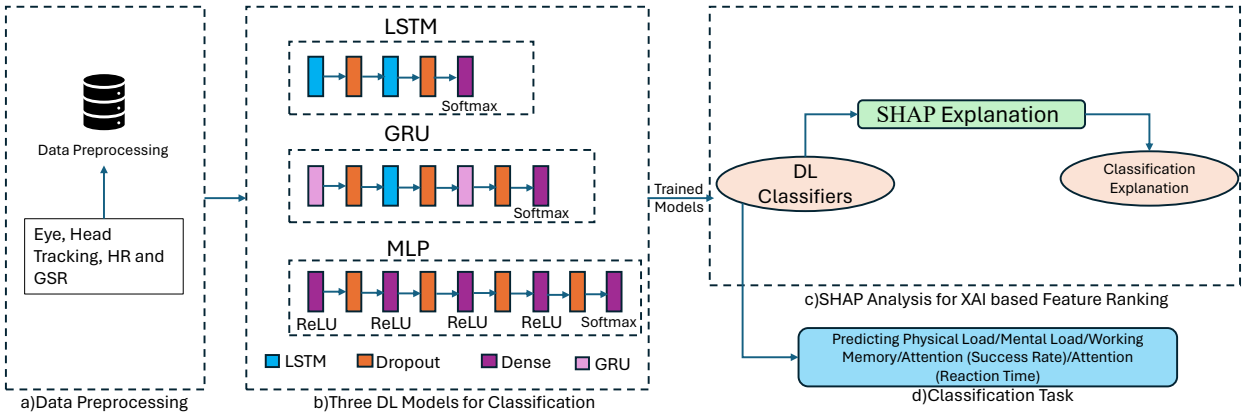


Fig. 1: Overview of research methods: a) Data preprocessing, Time series data preparation, splitting into Training and Testing set b) Three separate DL classifiers Trained on each label type separately c) SHAP analysis on the trained DL classifiers d) Classification Task using the trained DL classifiers

**Abstract**—As VR technology advances, the demand for multitasking within virtual environments escalates. Negotiating multiple tasks within the immersive virtual setting presents cognitive challenges, where users experience difficulty executing multiple concurrent tasks. This phenomenon highlights the importance of cognitive functions like attention and working memory, which are vital for navigating intricate virtual environments effectively. In addition to attention and working memory, assessing the extent of physical and mental strain induced by the virtual environment and the concurrent tasks performed by the participant is key. While previous research has focused on investigating factors influencing attention and working memory in virtual reality, more comprehensive approaches addressing the prediction of physical and mental strain alongside these cognitive aspects remain. This gap inspired our investigation, where we utilized an open dataset - VRWalking, which included eye and head tracking and physiological measures like heart rate(HR) and galvanic skin response(GSR). The VRwalking dataset has timestamped labeled data for physical and mental load, working memory, and attention metrics. In our investigation, we employed straightforward deep learning models to predict these labels, achieving noteworthy performance with 91%, 96%, 93%, and 91% accuracy in predicting physical load, mental load, working memory, and attention, respectively. Additionally, we conducted SHAP (SHapley Additive exPlanations) analysis to identify the most critical features driving these predictions. Our findings contribute to understanding the overall cognitive state of a participant and effective data collection practices for future researchers, as well as provide insights for virtual reality developers. Developers can utilize these predictive approaches to adaptively optimize user experience in real-time and minimize cognitive strain, ultimately enhancing the effectiveness and usability of virtual reality applications.

**Index Terms**—virtual reality, real walking, cognitive load, attention, working memory, machine learning, deep learning, classification, explainable ai

## 1 Introduction

Virtual reality (VR) finds widespread applications across various domains, many of which involve users engaging in multitasking scenarios demanding high precision and efficiency. Such applications necessitate individuals to concurrently juggle multiple tasks within the virtual environment. Effective performance in VR relies on maintaining attention and having sufficient working memory to handle complex tasks while navigating virtual environments. Thus, attention and cognition in VR

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*Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxxx*

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have become important areas of research in modern VR studies, aiming to enhance user performance and experience in immersive virtual environments. However, the combination of navigation, 3d interaction, and cognitive tasks may be taxing to users' cognitive load, attention, memory, and physical load. If the application could predict the users' cognitive and physical load, the application could adapt to the user in real-time, thereby optimizing the virtual experience to help the user meet their objectives. A gap in prediction research hinders developers from creating adaptive VR environments, which our research proposes to address.

To demonstrate the need for prediction in VR, consider, for instance, firefighter response training conducted through VR [65], where users must navigate a virtual building to locate survivors amidst simulated fires intended to expose users to sensory overload [66]. Trainees may be physically moving through the environment via real walking or crawling, which adds to the physical load. Trainees are looking and listening for survivors and maintaining a heightened awareness of the burning structure of the building, thereby adding to the demand on attention. Throughout this exercise, trainees must be able to remember instructions or information about the potential survivors that could influence plans for extraction, which involves working memory. Moreover, in a VR environment, the headset and controllers may also have a demand on cognitive and physical load, adding some potential unwanted contributions to overload. However, trainees with less experience may have lower capacities of cognitive load, attention, and working memory in such an environment, meaning they may experience better training outcomes if those demands could be lowered to meet their current level of training. If the virtual environment can make accurate predictions about the user's cognitive load, attention, and working memory load, it could adapt the scenario to be more effective for the user, for example, reducing the spread of fire or making survivors easier to attend to.

Recent work in prediction in VR has focused on predicting cybersickness via deep learning [28,45]. The common approach to predicting cybersickness is to train models with multimodal datasets (e.g., eye tracking, head tracking, physiological measures, images) labeled with a metric similar to the Fast Motion Sickness Scale. Prior works have increasingly demonstrated excellent performance in classification and regression tasks, for example, using a variety of basic and more complex models [25] and model optimization [33]. However, this prior research failed to take into account the effects on cognitive load, attention, and working memory.

For attention prediction in VR, most prior work was focused on predicting *visual attention*, which attempts to predict the saliency of objects within the virtual environment. Current state-of-the-art methods can predict the saliency of virtual objects at over 90% accuracy [18,35]. However, virtual object saliency is not a measure of how demanding a virtual environment is on a user's attention. Our proposed work is focused on predicting a user's attentional demand, which has more direct implications on task performance and learning [44].

For recent cognitive load prediction in VR, researchers use a method similar to the aforementioned cybersickness prediction through multimodal data collection (e.g., eye tracking, physiological signals, electroencephalogram) during a dual task. Sarkar et al. trained a classifier to predict cognitive load in an augmented reality-enabled medical patient simulation to 89% accuracy. However, they only collected a dataset of nine participants. [48]. Souchet et al. trained a classifier to predict three classes of cognitive load at 87% accuracy [53]. Our work specifically separates physical and mental load to obtain a more granular indication of cognitive load.

It is generally thought that cognitive load and attentional demand can have an effect on working memory load and vice versa [34]. However, to our knowledge, prior work has not attempted to predict working memory load in VR via deep learning.

To address these gaps, we developed, trained, and evaluated several deep-learning models (table 2) specifically to predict attention, cognitive load - separated into mental load and physical load - and working memory load during VR. We trained our models using the VRWalking dataset [51], which includes many types of input data, such as eye tracking, head tracking, heart rate, and galvanic skin response obtained while users executed multiple mental tasks while physically walking through mazes in VR. We chose to use this dataset because it was the only one we found that had labeled multi-task data collected throughout the VR experience, which was in line with our research questions about predicting attention, cognitive load, and physical load in VR navigation. Much of the prior work in the area of cognitive load prediction has used pre and post-questionnaires (e.g., NASA-TLX) to obtain this ground truth, which may not be sufficient to predict the dynamic changes in load during the experience effectively. Results suggest that attention, mental load, physical load, and working memory load can be predicted effectively at over 90% accuracy in most cases with the proposed approaches. Lastly, we present SHAP analyses for the highest-performing models to indicate the most impactful data features to use for such predictions in future models.

The proposed contributions of this research include:

- **Attention prediction:** We developed and evaluated models to predict attention based on reaction time and percentage of accuracy during a dual task.
- **Mental load prediction** As a more granular aspect of cognitive load, we developed and evaluated models to predict cognitive load based on a single question derived from NASA-TLX [61].
- **Physical load prediction** As a granular feature of cognitive load, we developed and evaluated deep learning models to predict physical load.
- **Working memory load prediction:** We leveraged task performance data from the digit recall task conducted during the session to train deep learning models for predicting working memory load.
- **Rigorous analysis of feature impact:** We conducted SHAP analyses for attention, physical load, mental load, and working memory load to investigate the impact of data features, which can be used in the future for training more advanced models.

This paper is organized as follows: First, we present the background literature on cognitive load, working memory, and attention, identifying the gaps in research around these areas concerning prediction in VR. Then, we describe our overall research questions related to this gap and our proposed contributions in predicting attention, mental loads, and physical loads. Next, we present our methodology, including a description of the dataset used, the pre-processing of the data, model architectures, and training setup. Then, the classification and explainability results are presented and discussed.

## 2 Background

In this section, we define cognitive load, attention, and working memory, discuss measurement and prediction methods, and highlight the research gaps at the intersection of these areas.

### 2.1 Cognitive Load

Cognitive Load encompasses both the mental and physical effort required to complete a task. Cognitive Load Theory (CLT) breaks down cognitive load into intrinsic load (related to task difficulty), extraneous load (involving mental demands), and germane load (associated with long-term memory) [58]. Much research on working memory has focused on reducing extraneous load, as adjusting instructional methods offers a practical means of reducing cognitive demands [49]. Previous studies have utilized the Paas Scale to gauge mental load [1, 57], which consists of a single 9-point Likert scale question rating mental load. However, as noted by Aldekhyl et al., NASA TLX provides a better measure for understanding the subtypes of cognitive load as defined by CLT [1]. In the VRWalking dataset, the researchers collected verbally reported measures similar to the Paas scale [38] during the navigation task and administered the NASA TLX afterward. While recent research predominantly employs the NASA TLX, administering a Paas scale-like measure during the session allows for more nuanced labeling of data. Previous research has been conducted on cognitive load prediction [20, 37]. However, they only utilized EEG signals for

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cognitive load prediction. The use of multimodal physiological data can also be seen in recent research [36]. However, utilizing comprehensive multimodal data containing head tracking and eye tracking alongside physiological measures would make the prediction model more robust. Also, it is important to note that very little prior research has focused on cognitive load during a real walking-based navigation multitasking in virtual reality [3].

To accurately characterize cognitive load and depict a participant's cognitive state, it's essential to consider not only the levels of physical and mental load but also factors such as working memory capacity and attentional resources.

### 2.2 Working Memory

Working memory, responsible for temporarily storing and manipulating information, has a limited capacity, handling only a modest amount of data, whether abstract ideas or simple numbers [5, 16]. To assess working memory, the authors of the VRWalking dataset used a method similar to the digit span test, where participants were given a set of 5 digits to remember and then asked to repeat them. Unlike the original test, they kept the length of the digit sequence constant throughout their test. Kim et al. employed a similar method involving both backward and forward digit span tests [29]. The impact of cybersickness, which can significantly affect cognitive and motor skills in individuals using virtual reality, has been well-documented [30]. In today's virtual reality applications, understanding how these applications affect users' mental workload and performance is important. Users also need to effectively manage their working memory in virtual environments, especially given the increasing complexity of tasks over time.

Although cognitive load theory often references Baddeley et al.'s working memory model [4], some researchers have noted inconsistencies in this connection [50]. Schuler et al. emphasized the importance of considering working memory as a control factor in studies. To address this, the authors included a memory task in their data collection to assess working memory based on task performance, taking into account the potential impact of multiple tasks and cybersickness.

In recent years, research relating to the working memory of participants immersed in virtual reality is becoming more and more popular [14, 62]. However, this research also fails to address the impact of navigation based on real walking and multitasking on working memory. Moreover, it is important to note that we found no research focused on predicting working memory in VR.

### 2.3 Attention

Attention refers to a set of processes that prioritize certain elements of the environment or tasks, aiding in the comprehension of information [2]. As attention is limited in capacity, individuals face challenges when attempting to perform multiple tasks simultaneously. In VR, studying human visual attention can involve utilizing readily available data from head-mounted displays (HMDs), such as eye-tracking, head-tracking, and stereoscopic video data [59], or employing dual task metrics [23]. However, there has been limited research investigating the relationship between attention and the effects of multiple tasks on attention, particularly when users are in motion.

Other considerations essential to our proposed work include selective Attention and divided Attention. Selective attention is tied to only one task, whereas attention can be divided into multiple tasks. The capacity of both selective and divided attention is limited. Previous work shows that participants are passively watching a video game and pay less attention to the game. They are also more susceptible to cybersickness than participants who are actively playing the game [6]. That may mean attention is correlated with cybersickness susceptibility. Moreover, recent studies in the field of VR have placed growing emphasis on predicting multimodal visual attention [18, 35, 60]. Attention metrics such as task-based attention and reaction time during virtual tasks are acknowledged as vital markers of attention in multitasking scenarios [19]. Nonetheless, the presence of labeled data for attention prediction in current datasets remains notably scarce.

But, the impact of multitasking on attention and vice versa is still not thoroughly studied in the VR research area. Omnidirectional video content or cinematic virtual reality with a wide field of view that is delivered through an HMD is technologically and interactively much more sophisticated than a video game. It also requires the user to focus on multiple tasks inside a VE. So, attention deficits can cause dual-task impairments and hinder the user experience while immersed inside a VE. More research is needed to understand the relationship between attention and multitasking.

### 2.4 Explainable AI (XAI)

Challenges like cybersickness and cognitive load remain significant barriers to VR usability. Recent advances in machine learning (ML) and explainable artificial intelligence (XAI) provide promising solutions for addressing these challenges. Eye-tracking metrics, such as saccades, fixations, and blinks, have been identified as key indicators of emotional and cognitive states in VR environments. Using datasets like VREED (VR Eyes: Emotions Dataset), studies have employed techniques such as SHapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) to reveal correlations between eye movements and emotional arousal, enhancing VR system adaptability [7]. To detect cybersickness, lightweight and interpretable ML models, such as LSTM and GRU, have been optimized using XAI methods to reduce computational complexity while maintaining accuracy, making them suitable for standalone VR headsets [33]. Super learner-based ensemble models have also been developed, combining multimodal features like eye tracking, galvanic skin response, and player position to classify cybersickness into levels with high precision, leveraging XAI-guided feature selection [31]. Additionally, physiological signals like heart rate variability (HRV) and electrodermal activity (EDA) have been integrated with machine learning to predict cognitive load and HUD complexity in VR and driving simulators. Feature fusion techniques have achieved high classification performance, emphasizing the potential of physiological data for real-time VR system optimization [55]. Finally, studies on spatial decision-making in VR environments highlight the influence of user expertise and reflexive motor skills on navigability, using SHAP to interpret neural network-based predictions [56]. Multi-modal approaches combining EEG and EDA signals further demonstrate the efficacy of feature selection for cognitive load classification, achieving high accuracy with models like XGBoost [9]. These advancements underline the role of XAI in enhancing the interpretability, efficiency, and adaptability of VR systems.

## 3 Dataset

The VRWalking dataset [51] was collected as follows. Participants navigated virtual mazes through real walking, a process lasting 15 minutes. Each minute slot has two divisions: the first 30 seconds were attention task, and the last 30 seconds was a Q&A session when the Likert scale questionnaires for cybersickness, physical load, and mental load were administered, and the working memory task response was recorded. Thus the participants faced the Q&A session every minute. Prior research has employed a consistent time window of 30 seconds to 1 minute for labeling datasets [25, 26, 33, 64]. The VRWalking dataset follows a similar methodology in generating its labels. The dataset includes data from 39 participants (23 males, 16 females) with an average age of 25.67 and a standard deviation of 7.22. Among the participants, 25 out of 39 had previous experience with VR equipment, while 13 had no prior VR experience. Additionally, the participants represent diverse backgrounds: 15 Hispanic, 3 African American, 8 Asian, 11 White, 1 Alaska Native, and 1 preferred not to disclose their ethnicity. The dataset contains eye tracking, head tracking, galvanic skin response, and heart rate data with ground truth data on cybersickness, working memory, attention, and cognitive load. The following contains more detail on the ground truth data that we utilized in our research.

*Attention:* The participants in the VR session heard one word at a time, chosen at random from a pool of five words ("Alpha", "Bravo", "Charlie", "Delta", and "Echo"), while they navigated the maze. This happened every two to three seconds. At the start of the session, each participant received a specific target word. They were required to skip if

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Table 1: Different Modalities of Data from the VRWalking Dataset

| Data Type | Feature Name |
|---|---|
| Subjective Information (Label Data) | Cybersickness, Physical Load, Mental Load, Working Memory, Attention (Success Rate), Attention (Reaction Time) |
| Eye Tracking Data | GazeTarget_X, GazeTarget_Y, GazeTarget_Z, EyeFrames, ViveTiming, Combine_Origin_X, Combine_Origin_Y, Combine_Origin_Z, Combine_GazeDir_X, Combine_GazeDir_Y, Combine_GazeDir_Z, Combine_Validity, Left_Origin_X, Left_Origin_Y, Left_Origin_Z, Left_GazeDir_X,..... 61 Total Features |
| Head Tracking Data | HMDVec_X, HMDVec_Y, HMDVec_Z, HMDPos_X, HMDPos_Y, HMDPos_Z, HMDRot_X, HMDRot_Y, HMDRot_Z, HMDRot_W |
| Physiological Data | Heart Rate (HR), Galvanic Skin Response (GSR) |

the word they heard matched the target word and to press the designated button when they heard the words during the session. To evaluate the participant's attention, two performance metrics were computed from this task [23].

- Accurate Button Presses: This is a percentage of the number of accurate button presses. This includes pressing the button correctly when hearing a word other than the target word and pressing the button correctly when skipping the target word. This is denoted as Attention (Success Rate).
- Reaction Time: Reaction Time is the duration between the audio cue played through the headset and the subsequent button presses. This duration is then averaged within one-minute intervals to correspond with physical load and mental load feedback collected every minute and synchronized with the collection of working memory data. This is denoted as Attention (Reaction Time).

*Working memory load:* Participants were presented with a five-digit number through headphones. During the Q&A session, they were asked to repeat the five digits, followed by receiving a new randomly generated five-digit number. Working memory load was assessed by calculating the percentile of correctly repeated digits. A digit was considered correct if both its index and value matched those of the given number. For instance, if the given number is 67849 and the participant responded with 68859, we interpret this as 6X8X9, X indicating missed digits. The performance score is determined accordingly, such as 60%. This task resembles the digit span test [10] utilized in previous studies [29].

*Physical Load:* This is a measure of the multitask's physical demand from a single question from the NASA-TLX [21] every minute: "How physically demanding is the task on a scale of not physically demanding/1-very high physical demand/10?" This was asked once per minute.

*Mental Load:* In line with the Paas scale [58] and NASA-TLX [21], this was a participant's response to: "How mentally demanding is the task on a scale of not mentally demanding/1-very high demand/10?" collected during the Q&A session. This was also asked once per minute.

## 3.1 Data Preprocessing

The dataset consisted of a total of 61 features from the eye-tracking data, encompassing measurements such as pupil diameter for both eyes, normalized gaze direction, and convergence distance. Additionally, the dataset had ten features for the head-tracking data, including values for head orientation quaternions (x, y, z, and w). To ensure comparability and consistency between the eye-tracking and head-tracking data, we normalized them using the following formula:

$$X_{\text{normalized}} = \frac{X - \mu}{\sigma} \tag{1}$$

Here, $\mu$ is the mean of the data, and $\sigma$ is the standard deviation of the data.

We utilized exponential smoothing to process the collected physiological readings, employing the following equation. This technique is valuable for uncovering the temporal dynamics present in heart rate (HR) and galvanic skin response (GSR) data. Since physiological data is noisy, exponential smoothing helps the models learn more easily.

$$\hat{y}_{t+1} = \alpha \cdot y_t + (1-\alpha) \cdot \hat{y}_t \tag{2}$$

Here, $\hat{y}_{t+1}$ is the forecasted value at time t+1, $\alpha$ is the smoothing parameter, a value between 0 and 1, $y_t$ is the actual value at time t, and $\hat{y}_t$ is the forecasted value at time t. We experimented with the $\alpha$ value and ultimately decided to set it to 0.8. We determined the $\alpha$ value for the exponential smoothing through a trial and error approach. Initially, we visualized the raw Heart Rate (HR) and Galvanic Skin Response (GSR) data, experimenting with various $\alpha$ values ranging from 0.1 onwards. After careful examination of the visualizations, we selected the optimal $\alpha$ value of 0.8 for the final exponential smoothing, as this value was effective in reducing the noise optimally without deviating too much from the original signal. A similar approach was also taken by recent research working with HR and GSR [25, 26].

### 3.1.1 Categorization

To support classification, we categorize the target values into three categories. Equation 1 illustrates the methods used for classification.

$$Class_t = \begin{cases} \text{Low}, & \text{if } 0 < X \leq Q_1 \\ \text{Medium}, & \text{if } Q_1 < X \leq Q_2 \\ \text{High}, & \text{if } X > Q_2 \end{cases} \tag{3}$$

In Equation 3, $X$ denotes the type, where $X$ can represent $PL_t$ (Physical Load), $ML_t$ (Mental Load), $WML_t$ (Working Memory Load), and $ATT_t$ (Success Rate). The quantiles were defined as follows: $Q_1 = 2$ and $Q_2 = 3$ for $PL_t$ and $ML_t$. This deliberate choice of $Q_1 = 2$ and $Q_2 = 3$ was made to effectively classify cognitive load while ensuring dataset balance. Despite the score range of $PL_t$ and $ML_t$ spanning from 1 to 10, selecting a narrower range aimed to achieve dataset balance, given that levels exceeding 6 were uncommon. Hence, choosing $Q_1$ and $Q_2$ was ideal for creating a well-balanced classification dataset. For $WML_t$ and $ATT_t$ (Success Rate), $Q_1 = 20$ and $Q_2 = 40$ were set, considering their range from 0 to 100.

$$ATT_t(ReactionTime) = \begin{cases} \text{Fast}, & \text{if } 0 < X \leq Q_1 \\ \text{Moderate}, & \text{if } Q_1 < X \leq Q_2 \\ \text{Slow}, & \text{if } X > Q_2 \end{cases} \tag{4}$$

As for $ATT_t$ (Reaction Time), quantile-based binning was utilized for dataset balance, where the quantiles were defined as [0, 1/3, 2/3,1] with $Q_1 = 0.33$ and $Q_2 = 0.66$, and the categories are as described in Equation 4. This defines the quantiles at which to split the data. Specifically, it uses the 0th, 33.33rd, 66.67th, and 100th percentiles. As the $ATT_t$ (Reaction Time) was scaled to be in 0 to 1 range the $Q_1 = 0.33$ and $Q_2 = 0.66$.

### 3.1.2 Time Series Data Preparation

On a scale of 1 to 10, users self-assessed their current sentiments about the physical load, mental load, and overall level of sickness at certain times, roughly once every minute, represented by the time t. Furthermore, we computed their memory and attention scores according to the VR tasks they were engaged in. Cybersickness has been linked to delayed reaction times [42,43]. Therefore, it's possible that participants' self-assessment ($PL_t$, $ML_t$) was recorded at a time that was not exactly when they started feeling uncomfortable. Here, $PL_t$, $ML_t$, $WML_t$, $ATT_t$ (Success Rate), and $ATT_t$ (Reaction Time) have been calculated using Equations (3) and (4). $PL_t$, $ML_t$, $WML_t$, $ATT_t$ (Success Rate), and $ATT_t$ (Reaction Time) represent the categorical Physical Load, Mental Load, Working Memory Load, Attention (Success Rate), and Attention (Reaction Time), respectively. We incorporated all accessible data, such as heart rate (HR), galvanic skin response (GSR), and eye and head tracking information, in a time window spanning from t minus 60 seconds(t - 60s) to t ($t_s$) in order to account for this potential temporal lag. Although $ATT_t$ (Success Rate) and $ATT_t$ (Reaction Time)

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can be calculated at a higher frequency from the timestamped log included in the VRWalking dataset, we opted to utilize a 60-second time frame. This decision was made to align with the frequency of other labels such as $PL_t$, $ML_t$, and $WML_t$. We purposefully selected a 60-second time limit to match the dataset generated from self-assessment questions at regular intervals, with Q&A sessions usually taking 30 seconds to complete. With the use of this dataset, we were able to better comprehend the participant experiences by creating data points of interest corresponding to $PL_t$, $ML_t$, $WML_t$, $ATT_t$ (Success Rate), and $ATT_t$ (Reaction Time).

### 3.2 Neural Network Models: LSTM, GRU & MLP

We utilized three distinct neural network architectures for severity classification: LSTM (Long Short-Term Memory), GRU (Gated Recurrent Unit), and MLP (Multilayer Perceptron). LSTM and GRU are recurrent neural network (RNN) architectures primarily developed for processing sequential data like time series. MLP, in contrast, is a feedforward neural network composed of interconnected layers of neurons, capable of handling time-series data as well. As depicted in Table 2, the LSTM model consists of two LSTM layers designed to capture sequential patterns, each followed by a dropout layer to prevent overfitting. The final dense layer enables multi-class classification. Similarly, the GRU model integrates a GRU layer to capture temporal dependencies, followed by a dropout layer for regularization. It also includes an LSTM layer for intricate temporal modeling, another dropout layer, and a second GRU layer with additional dropout. The final layer serves classification purposes. The MLP model begins with an initial dense layer for feature extraction, followed by dropout to ensure regularization. It comprises two additional hidden dense layers with a dropout layer between them, concluding with a dense layer for multi-class classification. Our primary preference for classification revolves around the use of simpler models. This strategic decision not only facilitates SHAP Analysis but also enhances interpretability. It is worth noting that applying SHAP analysis to complex models, like the deep fusion models employed for regression tasks [24], can be exceptionally challenging and often complicates the identification of dominant features [32, 33]. Table 2 provides an illustration of the model architecture adopted for severity classification.

### 3.3 Loss Function

For the classification tasks, we used categorical cross-entropy as the loss function.

$$L = -\sum_{i=0}^{K}(X * log(\widehat{X})) \tag{5}$$

In Equation 5, X represents the ground truth class of physical load, mental load, attention, and working memory load, and $\widehat{X}$ represents the predicted severity class and K is the number of classes.

### 3.4 Training Setup

We utilized a 10-fold cross-validation methodology to train and evaluate the performance of our proposed models, following established practices in prior studies [8, 39, 46]. In this approach, the dataset is divided into k=10 subsets, with one subset used for testing while the remaining (k-1) subsets are used for training [27]. This process is repeated k times, each time using a different subset for testing, which helps mitigate potential dataset bias [8].

To optimize model parameters during training, we reserved 30% of the training dataset as validation data for each fold [47]. We employed the Adam optimizer with 200 epochs of training and a batch size of 256, all configured through hyperparameter tuning. Additionally, to prevent overfitting, we implemented early stopping with a patience setting of 30 during model training [13].

## 4 Results

The following subsections go over the severity classification for predicting mental load, physical load, working memory load, attention(success rate), and attention(reaction time). The subsections also include the results of the explainable AI based SHAP analysis approach for learning the dominant features.

Table 2: Model Architectures for Classification

| Layer | Type | Output Shape | #Param | Dropout Rate | Activation |
|---|---|---|---|---|---|
| **Architecture for the LSTM Model** | | | | | |
| 1 | LSTM | 256 | 264192 | 0.2 | - |
| 2 | Dropout | 256 | 0 | 0.2 | - |
| 3 | LSTM | 128 | 197120 | 0.2 | - |
| 4 | Dropout | 128 | 0 | 0.2 | - |
| 5 | Dense | 3 | 387 | - | Softmax |
| Total No. of Params: 461,699 | | | | | |
| **Architecture for the GRU Model** | | | | | |
| 1 | GRU | 32 | 3360 | 0.2 | - |
| 2 | Dropout | 32 | 0 | 0.2 | - |
| 3 | LSTM | 64 | 18816 | 0.2 | - |
| 4 | Dropout | 64 | 0 | 0.2 | - |
| 5 | GRU | 128 | 74496 | 0.2 | - |
| 6 | Dropout | 128 | 0 | 0.2 | - |
| 7 | Dense | 3 | 387 | - | Softmax |
| Total No. of Params: 97,059 | | | | | |
| **Architecture for the MLP Model** | | | | | |
| 1 | Dense | 32 | 2368 | 0.25 | ReLU |
| 2 | Dropout | 32 | 0 | 0.25 | - |
| 3 | Dense | 128 | 4224 | 0.25 | ReLU |
| 4 | Dropout | 128 | 0 | 0.25 | - |
| 5 | Dense | 32 | 4128 | 0.25 | ReLU |
| 6 | Dropout | 32 | 0 | 0.25 | - |
| 7 | Dense | 8 | 264 | 0.25 | ReLU |
| 8 | Dropout | 8 | 0 | 0.25 | - |
| 9 | Dense | 3 | 27 | - | Softmax |
| Total No. of Params: 11,011 | | | | | |

### 4.1 Classification

The average accuracy, precision, and recall scores obtained through 10-fold cross-validation are presented in Table 3. In the classification of physical load, the GRU and LSTM models exhibited the highest accuracy. However, for mental load classification, the LSTM model surpassed the GRU model, achieving an accuracy of 96%. Conversely, in the classification of working memory and both attention metrics, the GRU model outperformed the LSTM model, achieving an accuracy of 93%.

### 4.2 SHAP Analysis

To delve deeper into the performance of the models and gain insights into their behavior, we employed SHapley Additive exPlanations (SHAP). Figures 2, 3, and 4 showcase the predominant features for each classification category.

SHAP serves as a post-hoc explanation technique, ranking input features to elucidate the predictions of deep learning (DL) models such as LSTM, GRU, and MLP, among others, for a given set of input samples encompassing eye and head tracking, HR, etc. The architecture of the DL models utilized is outlined in Table 2. However, Mangalathu et al. underscore the challenge of interpreting complex DL models through SHAP analysis [40]. Such models often harbor intricate internal representations and feature interactions, posing hurdles in providing clear and easily comprehensible explanations for model predictions using SHAP. Hence, we opted for simpler DL models akin to those utilized in our classification tasks. The global explanation yielded by the SHAP analysis elucidates the dominant features influencing the prediction of various classification types. Notably, the DL models that exhibited superior performance in the classification tasks are listed. Specifically, GRU demonstrated remarkable performance in attention, working memory load, and physical load classifications, whereas LSTM excelled in mental load classification.

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Table 3: Mean Accuracy, Precision and Recall of the 10-Fold Cross Validation on Classification for Different Models

| Type | Used Models | % Accuracy | % Precision | | | % Recall | | |
|---|---|---|---|---|---|---|---|---|
| | | | Low | Medium | High | Low | Medium | High |
| $PL_t$ | GRU | 91 | 98 | 90 | 75 | 91 | 91 | 90 |
| | LSTM | 91 | 95 | 90 | 82 | 93 | 91 | 85 |
| | MLP | 88 | 99 | 81 | 93 | 90 | 99 | 45 |
| $ML_t$ | GRU | 94 | 97 | 88 | 96 | 92 | 92 | 96 |
| | LSTM | 96 | 94 | 93 | 97 | 98 | 90 | 97 |
| | MLP | 92 | 96 | 78 | 98 | 87 | 95 | 92 |
| $WML_t$ | GRU | 93 | 70 | 74 | 95 | 65 | 55 | 98 |
| | LSTM | 91 | 63 | 63 | 94 | 53 | 55 | 97 |
| | MLP | 91 | 65 | 71 | 94 | 52 | 41 | 98 |
| $ATT_t$ (Success Rate) | GRU | 75 | 76 | 73 | 78 | 78 | 72 | 75 |
| | LSTM | 91 | 90 | 89 | 94 | 93 | 90 | 88 |
| | MLP | 76 | 79 | 75 | 76 | 78 | 78 | 72 |
| $ATT_t$ (Reaction Time) | | | Fast | Moderate | Slow | Fast | Moderate | Slow |
| | GRU | 73 | 73 | 71 | 77 | 74 | 68 | 78 |
| | LSTM | 81 | 81 | 79 | 84 | 82 | 79 | 83 |
| | MLP | 59 | 53 | 59 | 72 | 83 | 28 | 62 |

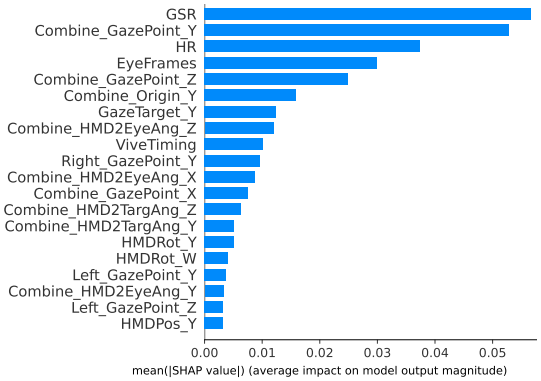


Fig. 2: Working Memory Load Classification (GRU)

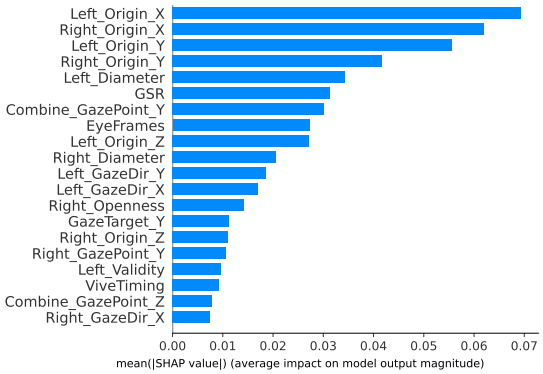


Fig. 3: Mental Load Classification (LSTM)

From the SHAP analysis, in addition to eye tracking, we can see that both HR and GSR were dominant features when predicting working memory and physical load. However, only GSR was prominent when predicting the mental load. Interestingly, head tracking had a dominant impact on attention(success rate), whereas eye tracking was more dominant in attention(reaction time).

#### 4.2.1 Global Explanation

SHAP-based global explanation methods provide a comprehensive understanding of the overall importance of features in machine learning models by aggregating SHAP values across an entire dataset. SHAP (SHapley Additive exPlanations) computes Shapley values, which quantify the contribution of each feature to individual predictions. For global explanations, SHAP calculates these values for all instances and averages them, offering a ranking of the most influential features in the model. In models like LSTM and GRU, which handle sequential data, SHAP captures the impact of time-dependent features, revealing how patterns over time contribute to predictions. For non-sequential models like MLP, SHAP evaluates how each individual feature affects the model's output, offering insights into non-linear relationships. See figures 2, 3, 4, 5, 6 for global explanation results, showing the features from top - most impact - to bottom - least impact.

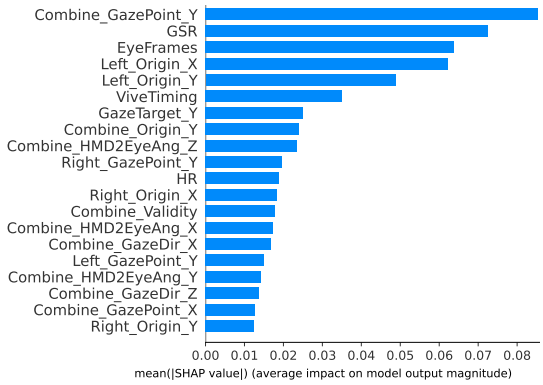


Fig. 4: Physical Load Classification (GRU)

#### 4.2.2 Local Explanation

SHAP-based local explanations focus on interpreting individual predictions by analyzing how each feature contributes to a specific instance's outcome. For each individual prediction, SHAP assigns Shapley values that quantify the contribution of every feature to the final decision,

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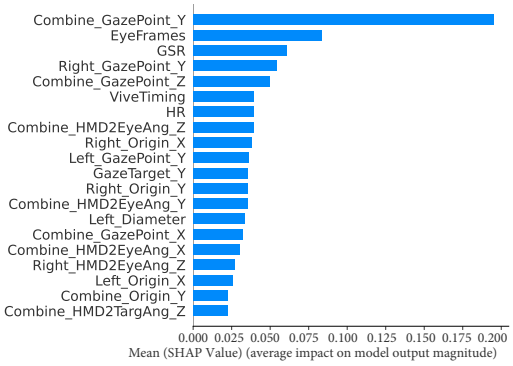


Fig. 5: Attention (Reaction Time) Classification (LSTM)

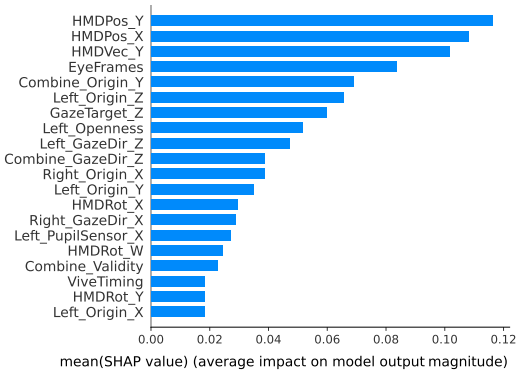


Fig. 6: Attention (Success Rate) Classification (LSTM)

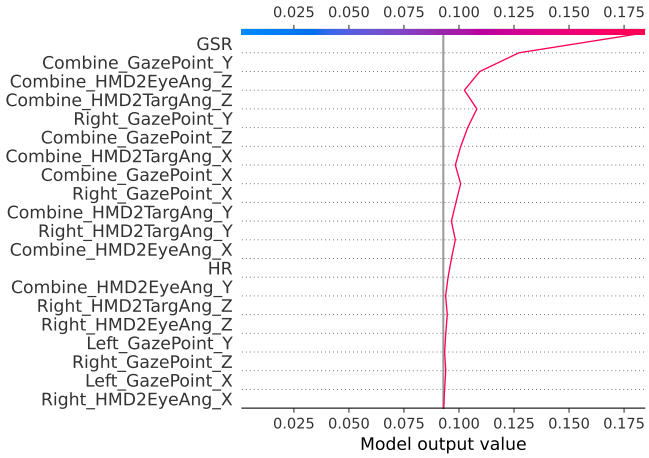


Fig. 7: Local Explanation Working Memory for High Class (GRU)

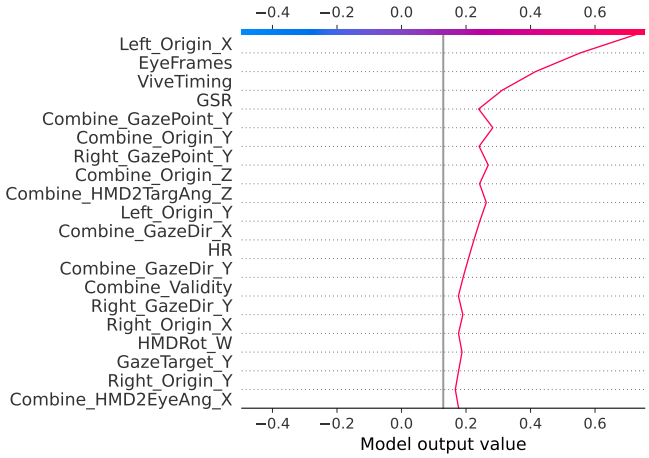


Fig. 8: Local Explanation Physical Load for High Class (GRU)

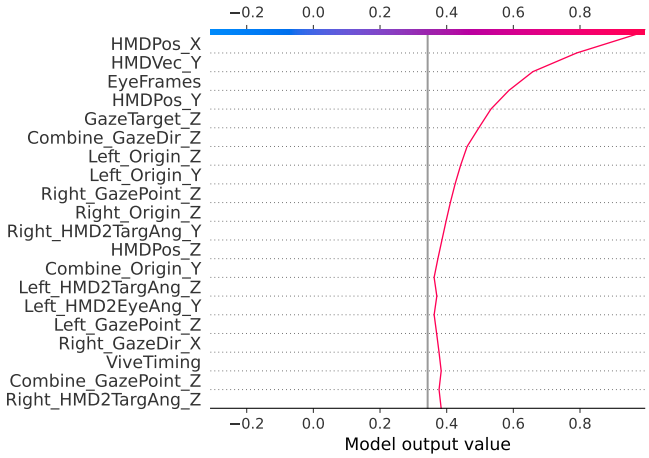


Fig. 9: Local Explanation Attention (Success Rate) for High Class (LSTM)

providing a clear breakdown of the factors influencing that specific prediction. This method allows for a detailed, case-by-case understanding of why the model made a particular prediction. Local explanations are particularly useful for identifying edge cases, uncovering model biases, or understanding variations in predictions for different instances. In sequential models like LSTM or GRU, local explanations reveal how specific time steps or sequences drive the prediction for an individual instance, while in models like MLP, they show the influence of individual features.

We analyzed the high categories of each feature type using SHAP local explanations to compare results with global explanations. The comparison reveals that while the ranking of the top features remains largely consistent, there are notable shifts. Specifically, the impact of HR decreased when predicting high working memory, indicating a reduced influence of HR in these cases as can be seen in Figure 7. In contrast, from figure 8 we can see that eye-tracking features and GSR showed diminished importance when predicting high physical load, whereas time series features representing the participant's duration within the simulation became more significant. This finding aligns with Setu et al. [51]'s observations that physical and mental load increase over time. Consequently, the importance rating of Eyeframes, which reflect timestamps, increased when predicting high mental load as can be seen in Figure 10. Upon examining Figure 9 we found out that for predictions of high success rate-based attention, the important feature space exhibited slight variations, though both global and local explanations consistently identified head-tracking features as the most critical.

## 5 Discussion

The aim of this study was to ascertain whether the overall cognitive state of participants, encompassing physical load, mental load, working memory load, and attention, could be predicted. Our findings indicate that leveraging straightforward deep learning models like LSTM, GRU,

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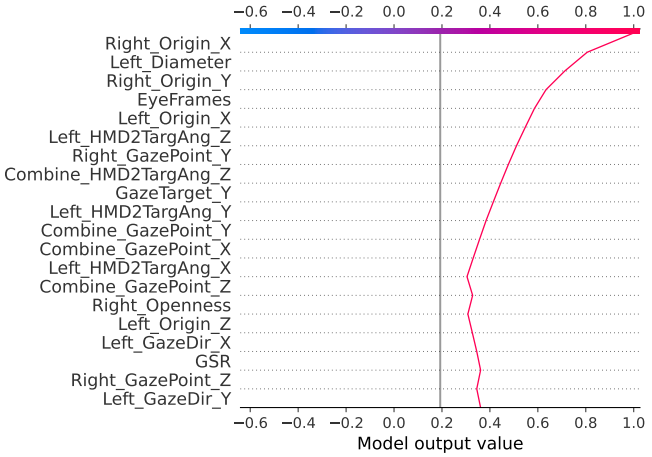


Fig. 10: Local Explanation Mental Load for High Class (LSTM)

and MLP, in conjunction with multimodal data (e.g., eye tracking, head tracking, HR, and GSR), facilitates the achievement of this goal. Notably, LSTM exhibited exceptional performance in timestamped data analysis, achieving accuracy of 91%, 96%, 91%, and 81% in predicting levels of physical load, mental load, task-based attention, and Attention (Reaction Time), respectively. However, GRU outperformed in predicting working memory load. Despite its inferior performance, the MLP model demonstrated the advantage of having fewer parameters, leading to reduced inference and training times. Notably, the GRU had 8.8 times more parameters, and the LSTM had 42 times more parameters than the MLP.

Using the same multimodal data, we can effectively predict mental load, physical load, working memory load, as well as both task-based and reaction-time-based attention. Together, these predictions provide a comprehensive view of a participant's overall cognitive state. For VR developers, this opens up new possibilities by revealing when VR simulations become physically and mentally challenging and when participants lose focus. This information enables developers to create more effective VR simulations that enhance the user experience.

### 5.1 Model Performance

The results presented in Table 3 demonstrate the performance of various neural network architectures, including GRU, LSTM, and MLP, across different tasks and evaluation metrics. Recurrent neural networks like GRU (Gated Recurrent Unit) and LSTM (Long Short-Term Memory) exhibit strong performance, particularly in tasks requiring temporal dependencies. GRUs generally excel in scenarios where computational efficiency is prioritized while maintaining high performance. For example, the GRU models achieve accuracy levels up to 94% in $ML_t$ tasks, with well-balanced precision and recall values across the low, medium, and high categories. This demonstrates the GRU's capacity for effective feature representation and classification within tasks involving temporal data. However, GRUs slightly underperform compared to LSTMs in some cases, such as the $ATT_t$ (Reaction Time) task, where temporal resolution and long-term dependencies might be more critical. LSTM models consistently deliver robust performance across all tasks, often outperforming GRUs and MLPs. For instance, the LSTM achieves 96% accuracy in the $ML_t$ task, with high precision and recall values across all categories. The architecture's ability to retain long-term dependencies while managing vanishing gradient problems makes it particularly effective for classification tasks involving sequential data. Similarly, in the $ATT_t$ (Reaction Time) task, the LSTM maintains higher accuracy (81%) and recall compared to GRU and MLP models. This highlights its suitability for tasks that require understanding of temporal patterns, such as reaction time prediction. The MLP (Multilayer Perceptron), a feedforward neural network, performs relatively well in some cases but struggles in tasks requiring temporal dependencies. For example, the MLP achieves 92% accuracy in the $ML_t$ task but exhibits lower precision and recall for medium and high categories compared to GRU and LSTM. In the $ATT_t$ (Reaction Time) task, the MLP's accuracy drops to 59%, reflecting its limitations in processing sequential information. This is expected since MLPs lack inherent mechanisms to model temporal relationships, relying solely on static feature representations. The varying performance across tasks and models underscores the importance of selecting architectures tailored to task requirements. Tasks like $ML_t$ and $PL_t$ involve classifying temporal patterns, which benefit from recurrent architectures like GRU and LSTM. In contrast, MLPs are more suited for tasks with static or spatially independent features, explaining their relatively lower performance in temporal tasks.

### 5.2 Working Memory Load Classification

As illustrated in Table 3, the fundamental DL models demonstrate remarkable performance in predicting working memory. Among the base models, GRU stands out as the top performer, achieving an accuracy of 93% for the working memory load classification task. In the context of working memory classification, we noticed the least precision (70%) in the 'low' class and the lowest recall (55%) in the 'medium' class. These findings imply that the model may generate false negatives for these two classes. Conversely, the elevated precision and recall in the 'high' class indicate that the model was successful in accurately predicting instances when working memory was at a higher level. While prior research has primarily concentrated on examining the factors influencing working memory within virtual reality [15, 41], our study took a different approach. We developed models to make predictions about the working memory of the study participants during the VR experience. Of note, we found no previous studies that focused specifically on predicting working memory. By successfully predicting working memory performance, developers can optimize task performance in virtual reality environments.

### 5.3 Cognitive Load Classification

We can see from Table 3, much like the success in predicting working memory, the fundamental DL models also exhibit strong performance in predicting and categorizing physical load and mental load. Specifically, both the base LSTM and GRU models achieved a notable accuracy of 91% in the classification task of physical load. For mental load, the LSTM surpassed the performance of the base GRU and MLP models, achieving an impressive accuracy of 96%. In both classifications, we noticed consistently high precision and recall across all classes. While the GRU model faced challenges in classifying the 'high' class, resulting in lower precision, the LSTM model effectively classified all classes. Unlike previous research [11, 54] that primarily focused on classifying cognitive load as one metric, our study achieved outstanding performance in predicting cognitive load at a more granular level that separated the physical and mental aspects.

### 5.4 Attention Prediction

Based on the obtained results, it is evident that different neural network architectures perform differently in predicting Attention (Success Rate) and Attention (Reaction Time). The performance of the models varied across the different architectures, including GRU, LSTM, and MLP.

For the $ATT_t$ (Success Rate), the LSTM model consistently outperformed the other architectures, achieving the highest overall accuracy (91%) and highest accuracy across all classes. This suggests that the LSTM architecture might be better suited for capturing the sequential patterns in the data related to $ATT_t$ (Success Rate). The GRU and MLP models also exhibited competitive performance but generally lagged behind the LSTM model in terms of accuracy.

Conversely, the performance variability observed in the prediction of Attention (Reaction Time) can be attributed to the relatively narrow range (Mean: $0.18 \pm 0.17$) of this metric compared to others, making it more challenging to distinguish between subtle differences. This restricted range reduces the spread of target values, leading to potential overlaps between predicted and actual classes. Additionally, models relying heavily on temporal features may struggle to capture the patterns necessary for precise prediction. This highlights the need for models

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with finer feature extraction capabilities, such as attention mechanisms, to better handle metrics with limited variability.

### 5.5 SHAP Analysis Discussion

The SHAP analysis provides deep insights into the underlying correlations and the features that impact model predictions (Figure 2, 3, 4, 5, 6). This information is invaluable for future data collection, enabling researchers to gather data more efficiently and develop lightweight prediction models. From Figure 2, we observe that for working memory load classification, physiological signals and eye-tracking features (e.g., Combine_Gazepoint_Y, Combine_Gazepoint_Z) have higher SHAP values, indicating a significant impact on the model's output. For mental load classification, only GSR and eye-tracking features like Left_Origin_X and Left_Origin_Y show substantial influence (Figure 3). This pattern is also evident in physical load and reaction-time-based attention classification (Figure 4, 5). However, head-tracking features (e.g., HMDPos_X, HMDPos_Y) become more influential when predicting task-based attention (Figure 6). The SHAP analysis uncovers intricate, deep-level correlations between features and severity types, enabling both VR researchers and developers to better understand the cognitive state and its relationship with various features. This understanding can lead to the development of more effective VR simulations and improved user experiences.

Eckstein et al.'s [17] work on eye tracking and cognition supports the SHAP analysis results. Eye-tracking measures offer non-invasive, rich indices of brain function and cognition, revealing attentional focus, cognitive strategies, mental effort through pupil dilation, and learning behaviors via blink rate. That is why we see Left_Diameter and Right_Diameter as some of the dominant features while predicting mental load. Similar work by He et al., Skaramagkas et al., and Wang et al. [22, 52, 63] also supports our findings by relating physiological and eye tracking data with cognition. Furthermore, as Figure 6 illustrates, head tracking becomes more dominant in the case of task-based reaction time, supporting the findings of Cai et al [12].

Moreover, the SHAP local explanation shown in Figure 7, 8, 9, 10 showcases the importance of time stamps and time-related features in predicting physical load, mental load and attention. It sheds light on the impact of the simulation's exposure time on participants. Although the global explanation does not deviate too much from the local explanation, interpreting the data from an instance level shows us slight to moderate differences in terms of feature rank. We chose to only present the results for high classes to properly understand which feature comes into play when the participant is at some extreme point in terms of cognitive load.

Furthermore, The SHAP analysis offers valuable insights into enhancing model scalability. By identifying the key subset of the dataset driving the prediction models, we can develop a lightweight model that operates effectively with less comprehensive data. Given that the data collection was conducted in a multitasking environment mirroring real-world scenarios, we anticipate that the model's accuracy will remain robust even in less controlled settings.

## 6 Limitations and Future Work

The VRWalking dataset used in this work has some limitations. Firstly, the dataset was not balanced in terms of gender. Datasets are more generalizable when the genders are balanced. This may have affected our prediction results. Secondly, the dataset's label distribution is skewed. For example, most of the data for physical load was in the 1-3 range rather than the total possible range of 1-10. This is the reason that we split the data by quantiles rather than attempting to predict over the entire 1-10 range. Thirdly, the Q&A frequency of 1 minute might have introduced additional noise into the dataset. However, a thorough comparison of intermittent questioning frequencies in VR data collection is lacking. Nonetheless, prior works have utilized this questioning method [25, 26, 64]. Lastly, the VRwalking data was collected concurrently in a multitasking scenario. This could mean that, for example, the tasks and measures for attention, cognitive load, and working memory could influence one another. Moreover, there could also be an effect of cybersickness as the participants spent 15 minutes in virtual reality. Future research is needed to produce datasets where each of these aspects is separated to better highlight causal relationships. However, we consider the multitasking nature of the dataset as an advantage for our work, as many virtual reality environments often include multitasking and may be subject to cybersickness, as exhibited by the motivating example of firefighter training [66]. Also, An integrated eye and head tracking system will facilitate the transfer of VR-based models to AR or MR environments. Further research is needed to determine the models' transferability to these other environments. Furthermore, SHAP-based explainable AI methodologies have been instrumental in constructing lightweight models by selectively retaining the most influential features and discarding the rest [33]. While this technique has primarily been applied to cybersickness prediction, its applicability extends to cognitive state prediction as well.

## 7 Conclusion

In conclusion, this study highlights the efficacy of multimodal prediction in determining the cognitive state of participants. We leveraged the VRWalking dataset, encompassing data on physical load, mental load, and task performance logs for the working memory and attention tasks. By using timestamped logs and verbally reported measures of physical and mental load, we constructed labeled data for training the deep learning models. Prior to training, we denoised the HR and GSR and normalized the eye tracking and head tracking data. This approach yielded significant performance in predicting the levels of physical load, mental load, working memory load, and attention. This approach also exemplifies a holistic method for predicting cognitive states, addressing physical load, mental load, working memory load, and attention, thereby bridging a gap in research for comprehensive cognitive state analysis. The utilization of SHAP analysis empowers future VR researchers to conduct more effective data collection, enriching datasets and enhancing their generalizability. Additionally, understanding the cognitive state of participants will enable VR developers to create environments that optimize user experience, leading to more immersive and efficient utilization of virtual environments.

## Acknowledgments

This work was supported in part by Army Research Lab awards W911NF2310401 and W911NF2420035, and National Science Foundation awards 2211785 and 2316240.

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This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2025.3549850

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